

resume of Oahidul Islam

Research Assistant
DeepHealth Research Lab
Sydney, Australia (Remote)

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EDUCATION

Daffodil International University (DIU)

Bachelor of Science (B.Sc.) in Electrical & Electronic Engineering (EEE)

(January 2019 - May 2023)

Major: Communication & Signal Processing (CSP)

CGPA: 3.97/4.00

Relevant Coursework: Power System | Power Electronics | Continuous Signals and Linear Systems | Digital Signal Processing I | Random Signals and Processes | Control System I | Wireless Technology | Probability and Statistics | AI & Machine Learning | Robotics

RESEARCH INTEREST

Applied Machine Learning & Deep Learning | Wireless Technology | Agentic AI | large language models (LLMs) | Computer Vision | Conversational AI | Quantum AI/ Computing | Signal Processing | Generative AI

RESEARCH EXPERIENCE

An explainable AI-based blood cell classification using optimized convolutional neural network

Principal Investigator: *Dr. Md Zahid Hasan*

- The primary objective of this research endeavor is to develop an enhanced CNN model for precise and efficient white blood cell (WBC) classification, achieving a testing accuracy of 99.12%. By employing various image pre-processing techniques and optimizing architectural structures and hyperparameters, our model outperforms existing transfer learning models. To improve interpretability, we utilized SHAP, LIME, Grad-CAM, and Grad-CAM++ techniques, providing valuable insights into model decisions.

PUBLICATIONS

Title: An explainable AI-based blood cell classification using optimized convolutional neural network

Authors: Oahidul Islam*, Md. Assaduzzaman*, Dr. Md. Zahid Hasan [*Equal Contribution] **Journal Article**

ScienceDirect- Journal of Pathology Informatics-Volume 15

December 2024, <https://doi.org/10.1016/j.jpi.2024.100389>

Title: Multi-Head Self-Attention Mechanisms in Vision Transformers for Retinal Image Classification COMPAS

Authors: Oahidul Islam*, Kowshik kumer, Sumaia Akter*, Md Mohi Uddin [*Equal Contribution] **Conference Paper** IEEE

CONFERENCE ON COMPUTING APPLICATIONS AND SYSTEMS (COMPAS 2024), CUET, Bangladesh, September 2024.

Title: Tuberculosis Disease Detection from Chest X-rays Using Deep Learning Techniques

Authors: Mehedi Hasan Rabby*, Oahidul Islam*, Md. Assaduzzaman*, Monoronjon Dutta [*Equal Contribution]

Conference Paper 2023 26th International Conference on Computer and Information Technology (ICCIT)- Cox's Bazar, Bangladesh, December 2023, [10.1109/ICCIT60459.2023.10441031](https://doi.org/10.1109/ICCIT60459.2023.10441031)

PROFESSIONAL EXPERIENCE

DeepHealth Research Lab | Sydney, Australia (Remote)

(Feb'25 – Present)

Research Assistant | XAI, Deep Learning, Gen AI

- Conduct research, writing manuscript
- Image preprocessing, segmentation, model analysis
- Deploy as webapp, android application, experimentation

Felicity Internet Data Center | Dhaka, Bangladesh

(December'24 – Feb-'26)

Engineer | Operation and Maintenance of Power System

- Maintenance of Power Equipment's
- Data analysis and report making

- Developed Model, Implementation Fine Tuning & Project Documentation.
 - Training and development of **Medical Image Segmentation, Computer Vision, Signal Processing Data.**
 - Development Content for [aionlinecourse](#) . Make documents, did research and implementation.
 - Developed real world projects to better understanding about AI, Machine Learning, Deep Learning for new Comers.

- Developed [Felicity-IDC.ai](#).
 - Information Extractor from documents
 - Developed Exploratory Data Analysis with Generative AI QnA.
 - Chatbot Integration: Query Felicity IDC's systems using an AI-powered chatbot.
 - Chat with multi documents Web-App using **RAG** and **Generative AI**.
 - Collaborate work with Electrical power, HAVC and BMS team. .

COMPETITIONS & ACHIEVEMENTS

- | | |
|--|------|
| • Global AI Agents League (Ongoing), Fetch.ai Innovation Lab
Building Finance AI Agent for financial analysis, stock market investment and so on. | 2025 |
| • Top 3% in World, Google AI Hackathon-2024 by Google
Ranked among the Top 3% finalists in the world’s largest AI hackathon. | 2024 |
| • Champion, DIU IoT Fest 2023
IoT project competition arranged by Dept. of CSE Daffodil International University. | 2023 |
| • Champion, 4IR - GrameenPhone.
Online competition arranged by Cisco & GrameenPhone. | 2022 |
| • Champion, Cyber Security Awareness Project Showcasing | 2022 |
| • 1 st Runner Up Robot Nokshar Ashor
Design Robot and become 1 st runners up in Bangladesh. Organized by Daffodil Robotics Lab | 2021 |

GRANTS AND SCHOLARSHIPS

- | | |
|--|-----------|
| - Recipient, Dean’s List Scholarship
Academic Honor by DIU for attaining CGPA 4 out of 4 for seven consecutive terms. | 2020-2022 |
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TECHNICAL SKILLS

Programming Languages and Development: Python***
Deep Learning Frameworks: TensorFlow*** | PyTorch | Computer Vision| Huggingface Transformers
Cloud Services and Frameworks: Amazon EC2 Services | Mircrosoft Azure | Google Cloud.
Database: MySQL | Vector Database, **Version Control:** Git | GitHub | Docker*, **Pipeline:** CI/CD, ETL, DagsHub*, Mlflow*
Numerical Analysis and Simulation Environment: MATLAB | MATLAB Simulink | PSICE | Proteus | Arduino IDE
Documentation and Illustration: LaTeX | AutoCAD

HIGHLIGHTED PROJECTS

- **AmarCare** ([GitHub](#))
Using Flask and multiple ML model for disease prediction
User advice/chat bot integration to assist using LLM
Real time Hospital/Doctor suggestion using **RAG+ Knowledge Graph**
- **Care AI** ([Web](#))
AI agents and language models to deliver a proactive healthcare experience. ([URL](#))

- **AI-Doctor-Alert** ([GitHub](#))
An AI-powered medical voice monitoring system that converts patient audio into text, analyzes medical urgency, and sends real-time alerts to doctors.
- **Research_kori – The Multi-Agent AI**
Multi agent ai works for research on Business, Company growth, Academic, Finance [\[GitHub\]](#)
- **Blood Cell Identification Using Android & Web APP** [\[GitHub\]](#)
Create real time identification for doctor and healthcare professionals.
- **DreamJob.ai** [\[Video Demonstration\]](#)
Created a comprehensive platform that leverages AI and machine learning to provide personalized career guidance.
- **Cervical Cancer Risk Factor analysis**
End-to-End MLOps with **AWS S3 buckets** to store model artifacts during model tracking and experimenting with **mlflow**.
- **SnappyText V1.0.1** [\[Amazon\]](#)
ML Based Image to Text Converter App. Build with **ML toolkit** and JavaScript. Support on Android v5.0+
- **SQL.AI:** [\[GitHub\]](#)
Text-To-SQL Code. The application connects to SQLite database, executes SQL queries, and provides visualizations of the retrieved data.
- **Car Tracking and Extract number plates with OCR** [\[Repository\]](#)
Detection of vehicles in traffic footage and count velocity using YOLO V12 based image processing system.
- **Face Attendance WEB APP** [\[Repository\]](#)
Real time image attendance system and save it on local directory as CSV. As well as, data analysis of attendance can also perform here.

MOOC COURSEWORKS

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| • Udemy offered course Generative AI Mastery. | March 2024 |
| • Deeplearning.ai offered course 'Generative AI with LLMs'. | April 2023 |
| • IBM offered course Machine Learning with Python'. | September 2022 |
| • Deeplearning.ai offered course 'Machine Learning in Production'. | August 2021 |

LEADERSHIP ACTIVITIES

CTO, Team Ape	(2022 - Present)
Vice-Captain, DIU EEE Cricket Team b-29 th	(2021 - 2023)
Executive Member, DIU EEE Club	(2020 - 2023)

CONTRIBUTOR

TOP Voice LinkedIn-Deep Learning	(2024 - Present)
Kaggle	
StackOverflow Streamlit TensorFlow GitHub Buy me a coffee Topmate	

REFERENCES

Kanij Ahmed

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Daffodil International University
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Md. Assaduzzaman

Assistant Professor, Dept. of CSE
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